Measurement Work Group

A broad goal for future research in measurement is to identify a set of procedures that can be used to help teachers make instructional decisions that improve student results. A hybrid system is needed as no one system is likely to be adequate for screening, progress monitoring, program evaluation, and instructional planning.

Past

The purposes of assessments/measurement have focused more on evidential validity and less on consequential validity.

Reading has gotten the most attention, with less emphasis on writing and mathematics; reading for primary age students (K-3) has been the most studied.

Skills analysis and task analysis skills that have been part of past could inform the development of tools for assessment. Teachers collected and used data much more for instructional planning purposes (MM within defined curriculum).

In existing studies, short term assessments can be used to identify the effectiveness of short-term interventions, may short-term change not be evident on long term assessments.
Some assessments (that have been abandoned) might need to be revisited (e.g., IRIs).

Present

We have many **different goals for** the application and use of measurement. Confusion and misconception (and under-researched) regarding aspects of GOM versus Mastery Measurements type of assessments; if MM are to be used, how can they be used effectively and appropriately (and with what psychometrics/technical adequacy)?

Within special education settings, some use of progress monitoring is occurring.

Research should focus on link to outcomes (that matter or are viewed as important).

Existing research does include some evidence and some syntheses (e.g., Jenkins – screening). Resources such as NCRTI tools charts have done some of the work in identifying existing and available materials. Commercially-available assessments/commercial products (e.g. MAP, Renaissance Star) exist but may not have been evaluated for consequential validity (or that these findings have not been widely shared) To what extent is it helpful for teachers to have access to these data? Are these measures sensitive to change?

Interventions and sensitivity to growth are inter-twined; an important topic/area for research is how monitoring progress within interventions can occur, including in evidence based interventions that are likely to be adopted. Are there assessments that work better for one intervention over another? How can you make decisions to choose or identify procedures that could be used for PM within a specific intervention? Current efforts focusing on how to use slope within effective instruction (e.g., how many points of a slope are needed) inform some use of slope as a criterion. Use of pre-post and slope debate continues/closely aligned and GOM systems may yield different findings/decisions about progress.

Future

Goal = to support improved outcomes for students; emphasis on the pragmatic application of tools that can be used for a variety of purposes

RESEARCH GOAL #1
Use existing literature (meta-analysis, review of lit) to identify LF's stages of CBM research x content x grade level x purpose and existing materials (e.g., task analysis).

RESEARCH GOAL #2
Identify the utility of the decisions that can be made based on the existing tools and how new tools can be developed or applied (and look for new relations among tools).

Outcomes need to be more clearly defined and consensus reached as to the educational outcomes that can/should be included in discussion (for different kinds of students and in different kinds of communities); outside of education resources might be helpful in structuring and defining these outcomes

Within Stage 1 and 2 areas of research, additional efforts are needed in these areas:

- reading for older students
- math,
- written language,
- content area at the middle and high school level (including content area literacy, content area knowledge/skills),
- preschool, and
- social/behavioral measures.

Stage 3 research for almost all areas/grade levels needs to be conducted. Predictive studies that might be able to better identify who is likely to experience later difficulties are needed.

The relationship between measurement and the treatment (i.e., instruction, intervention) is important.

Efficiency is key – using technology and other resources would be ways to address some of the efficiency concerns; question – what is the impact of assessment demands on instructional effectiveness? How much assessment is needed to make appropriate decisions? An additional method for developing (to be applied by teachers/school psych) assessments development process.

Reliability of slope takes more points than may be efficient. RQ: What is the consequential validity of getting a reliable slope estimate? What kinds of slope data are needed for different kinds of decisions about student progress? Growth is assumed to be linear – procedures exist that can be used to estimate nonlinear growth